Pakistan Flood Intelligence Dataset

Feature Engineering Report

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1. Dataset Overview

The **Pakistan Flood Intelligence Dataset** contains **13,742 district-day observations** spanning 52 districts across 6 provinces/regions (Punjab, Sindh, KPK, Balochistan, AJK, Gilgit-Baltistan) from **January 2000 to December 2025**. Each row represents conditions at a district centroid on a given day, encoded across **57 features** covering meteorological, hydrological, remote sensing, socio-economic, and impact variables. The dataset captures all major Pakistan flood seasons including the catastrophic **2010** and **2022** super-floods.

2. Feature Categories

Category	# Features	Variables
Temporal	6	date, year, month, day_of_year, season, monsoon_indicator
Geographic	7	province, district, lat, lon, elevation, basin, river_system
Meteorological	8	rainfall (daily/3d/7d), tmax, tmin, humidity, wind_speed, soil_moisture
Hydrological	5	discharge, water_level, warning_stage, reservoir_level, streamflow_anomaly
Remote Sensing	6	flood_extent, NDVI, NDWI, SAR_inundation, surface_water_occ, land_cover
Exposure	5	pop_density, built_up%, agri%, road_density, infrastructure_count
Engineered	7	rainfall_anomaly, ARI, return_period, dQ/dt, lag1/3/7_rainfall
Target/Event	4	flood_occurrence (binary), flood_type, is_glof, warning_stage_label
Impact	7	deaths, injured, displaced, houses_damaged, cropland, livestock, econ_loss
Metadata	2	data_source, quality_flag

3. Engineered Features — Design Rationale

Feature	Formula / Method	ML Rationale
rainfall_anomaly_mm	daily_rain – long_term_daily_mean	Detects anomalous precipitation events relative to climatology
antecedent_rainfall_index (ARI)	rainfall_7d × 0.7 + rainfall_3d × 0.3	Captures soil saturation memory; exponential decay weighting
rain_return_period_yr	(daily_rain/20)^1.5 if rain>20mm	Rarity signal; helps models identify extreme precipitation
discharge_change_rate_m3s	Q_t – Q_{t-1}	Rate of change captures sudden river rises (flash flood signature)
lag1_rainfall_mm	Rainfall at t–1 day	Captures immediate antecedent conditions
lag3_rainfall_mm	3-day average lagged rainfall	Medium-term antecedent moisture for riverine flood prediction
lag7_rainfall_mm	7-day average lagged rainfall	Captures basin-level saturation for large rivers (Indus, Chenab)
streamflow_anomaly_pct	(Q / Q_clim – 1) × 100	Relative discharge deviation from long-term climatological mean
monsoon_indicator	1 if month ∈ {Jun,Jul,Aug,Sep}	Binary season flag; most critical predictor in tree models

4. Target Variable Analysis

flood_occurrence (binary 0/1) is the primary target. Auxiliary targets available for multi-task learning:

Year	Flood Rate	Event Context
2022	86.8%	MEGA-FLOOD
2010	78.2%	MEGA-FLOOD
2020	74.5%	Major flood
2011	69.2%	Major flood
2007	57.7%	Major flood
2024	33.7%	Significant flood
2025	31.5%	Significant flood
2003	31.5%	Significant flood

Auxiliary Targets:

Target	Type	Description
flood_type	Multi-class	Riverine, Flash, Urban, GLOF, None
flood_warning_stage	Ordinal (0-3)	Normal, Watch, Warning, Danger
deaths	Regression	Casualty prediction
displaced_persons	Regression	Humanitarian impact estimation
economic_loss_usd	Regression	Financial damage assessment

5. Feature Importance & Correlation Guidance

Based on domain knowledge and expected statistical patterns:

Rank	Feature	Expected Importance	Domain Justification
1	rainfall_3day_mm	Very High	3-day accumulation is the strongest riverine flood trigger
2	monsoon_indicator	Very High	80%+ of Pakistan floods occur Jun-Sep
3	antecedent_rainfall_index	High	Soil saturation pre-conditions flooding
4	river_discharge_m3s	High	Direct flood indicator from gauges
5	water_level_m	High	Derived from discharge; proximal to inundation
6	streamflow_anomaly_pct	High	Captures unusually high flows vs climatology
7	elevation_m	High	Low-elevation districts (Sindh, Balochistan plains) flood more
8	flood_warning_stage	Medium	Lagged official warning state
9	vulnerability (encoded via district)	Medium	Historical district-level exposure
10	ndwi	Medium	Pre-flood surface water increase detectable

6. Preprocessing Pipeline Recommendations

Step	Method	Target Models	Notes
Missing value imputation	Median for numerics; 'Unknown' for categoricals	All	SAR/economic cols have 20-28% missing — use indicator variab
Categorical encoding	Label encoding for tree models; OHE for neural nets	All	district, province, flood_type, land_cover
Feature scaling	StandardScaler or MinMaxScaler	LSTM, CNN, Transformer	Not needed for XGBoost/LightGBM/RF
Class weighting	scale_pos_weight = n_neg/n_pos	XGBoost, LightGBM	Current overall flood rate ~32%; adjust per split
Sequence preparation	Sliding window (7-30 day lookback) per district	LSTM, CNN, Transformer	Sort by district+date; pad at district boundaries
Feature selection	Remove high-correlation pairs (r>0.95)	All	discharge_m3s ↔ water_level_m; handle multicollinearity
Temporal cross-validation	TimeSeriesSplit or year-blocked CV	All	Never use random K-fold on temporal data
Outlier handling	Winsorize at 99th percentile	Neural nets	Extreme discharge values in 2010/2022

7. Anti-Leakage Checklist

✓ Temporal split: Train ≤2019 | Val 2020-2022 | Test ≥2023 — no future data in training

✓ No leakage features: flood_extent, sar_inundation, deaths, displaced are IMPACT features — exclude from input, use as targets only

✓ Lag features computed correctly: lag1 = t–1 day; never use t+1 or t+2

✓ District-level group CV: when using cross-validation, group by year to prevent temporal leakage

✓ Impact columns (deaths, displaced, etc.) should ONLY be used as targets, not as input features

✓ 2010 and 2022 extreme events are in Train and Val respectively — model must generalise across magnitudes

✓ For LSTM/Transformer: use causal attention and no look-ahead masking